

LECTURE 5

Structural Breaks, State Space and the Kalman Filter

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Agenda

- 01 Types of structural breaks
- 02 Effect of unmodelled breaks on inference
- 03 Chow test and CUSUM procedures
- 04 Bai–Perron multiple break tests
- 05 Impulse Indicator Saturation (IIS)
- 06 State space representation
- 07 The Kalman filter
- 08 Time-varying parameter models and applications

(i) Brazilian inflation 1900–2018: multiple location shifts across political regimes. (ii) UK money demand: structural break at deregulation (1985). (iii) Time-varying CAPM beta.

Section 1

Types of Structural Breaks

A Taxonomy of Breaks I

Transitional breaks

Short-lived deviations from the baseline, captured by blip or impulse dummies:

$$D_t^{\text{blip}} = 1_{(t=t^*)} - 1_{(t=t^*+1)}$$

$$D_t^{\text{imp}} = 1_{(t=t^*)}$$

Examples: policy announcements, short financial crises, measurement errors.

Location shifts

Permanent change in the level (mean) of the series from date t^* onwards:

$$D_t^{\text{loc}} = 1_{(t \geq t^*)}$$

Examples: regime changes, structural policy reforms, central bank target changes. Brazilian inflation across governments: 9 distinct level regimes (1900–2018).

Levels and dynamics shifts

Both intercept **and** slope change at t^* :

$$Y_t = \alpha_1(t) Y_{t-1} + \mu(t) + \varepsilon_t$$

where α_1 and μ each shift at t^* .

Examples: currency peg collapses, financial crises that alter persistence of financial returns.

A Taxonomy of Breaks II

Why unmodelled breaks are dangerous

If the true DGP is $Y_t = \alpha Y_{t-1} + \mu_1 1_{(t \leq \tau_1)} + \mu_2 1_{(t > \tau_1)} + \varepsilon_t$ with $|\alpha| < 1$, but we estimate $Y_t = \rho Y_{t-1} + v + \varepsilon_t$, the OLS estimate $\hat{\rho}$ is biased **towards unity**.

An unmodelled location shift at $t^* = T/2$ with magnitude δ inflates $\hat{\rho}$ approximately by $O(\delta^2/\sigma^2)$.

Consequences for unit root testing

- ADF tests lose power against stationary alternatives with breaks
- May falsely conclude $I(1)$ when series is TS with a break
- Perron (1989): most Nelson–Plosser series are **stationary** once breaks are modelled
- Zivot–Andrews (1992): endogenous break detection in unit root test

Consequences for forecasting

- Location shifts at the forecast origin are the **primary** cause of forecast failure (Clements & Hendry, 1999)
- Standard forecasts fail to adapt to the new level
- Intercept correction and differencing provide robustness (covered in Lecture 7)

Section 2

Testing for Structural Breaks

Chow Test and CUSUM Procedures I

Chow (1960) test — known break point t^*

Estimate the model separately for $t \leq t^*$ and $t > t^*$. Test equality of all coefficients:

$$H_0 : \beta_1 = \beta_2 \quad \text{vs.} \quad H_1 : \beta_1 \neq \beta_2$$

F -statistic: $(RSS_R - RSS_U)/k \div RSS_U/(T - 2k) \sim F(k, T - 2k)$ under H_0 .

Requires the break date t^* to be known in advance. In practice, break dates are unknown and must be estimated — this leads to the Andrews sup- F test.

CUSUM and CUSUMSQ (Brown, Durbin & Evans, 1975)

Based on **recursive residuals** $\tilde{v}_t = (y_t - \mathbf{x}'_t \hat{\beta}_{t-1})/\sqrt{f_t}$ where $f_t = 1 + \mathbf{x}'_t (X'_{t-1} X_{t-1})^{-1} \mathbf{x}_t$.

- **CUSUM**: $W_t = \sum_{s=k+1}^t \tilde{v}_s / \hat{\sigma}$ — detects breaks in the **mean**
- **CUSUMSQ**: $S_t = \sum_s \tilde{v}_s^2 / \sum_{\text{all}} \tilde{v}_s^2$ — detects breaks in the **variance**
- Reject H_0 (parameter stability) if W_t or S_t crosses tabulated boundary lines

Chow Test and CUSUM Procedures II

Andrews (1993) $\sup$ - F test — unknown break point

Compute the Chow F -statistic for every break date t^* in the **trimmed** range $[0.15T, 0.85T]$:

$$\sup_{t^*/T \in [0.15, 0.85]} F(t^*)$$

Critical values from Andrews (1993) depend on the trimming fraction and number of parameters.

Extend Andrews to allow m break points $t_1^* < t_2^* < \dots < t_m^*$:

- 1 **Sequential testing**: test $m - 1$ vs. m breaks
- 2 **Global minimisation**: find the m break dates that minimise the total sum of squared residuals
- 3 **Dynamic programming**: computationally efficient algorithm
- 4 Critical values account for multiple testing

Estimated break dates are super-consistent: converge at rate T .

Section 3

Impulse Indicator Saturation

IIS and Step Indicator Saturation I

Add an impulse dummy $1_{(t=j)}$ for **every observation** $j = 1, \dots, T$ to the general unrestricted model. Since $N = T$ regressors $>$ observations, use the **Autometrics** algorithm (Lecture 7) to select which dummies to retain.

Properties of IIS

- Detects outliers and transitional breaks throughout the sample
- Works toward the very end of the sample (unlike Andrews sup- F)
- Gauge (false detection rate) is controlled by the target significance level α
- Gauge $= \alpha$ asymptotically (Johansen & Nielsen, 2016)
- Can be used with IIS + SIS together for comprehensive break detection

Step Indicator Saturation (SIS)

Add a location shift dummy $1_{(t \geq j)}$ for every observation. Designed to detect permanent level shifts rather than blips.

Combined IIS + SIS: very flexible — detects both transient and permanent structural changes. The basis of the **Autometrics** model selection algorithm.

Section 4

State Space and the Kalman Filter

State Space Representation I

Observation equation:

$$y_t = Z_t \alpha_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H_t)$$

State transition equation:

$$\alpha_{t+1} = T_t \alpha_t + R_t \eta_t, \quad \eta_t \sim N(0, Q_t)$$

α_t : unobserved state vector; Z_t, T_t, R_t : system matrices (can be time-varying).

Many models are state space

- ARIMA: state = lagged values
- Structural time series: trend + seasonal + cycle
- Time-varying parameter (TVP) regression
- Dynamic factor models
- Kalman smoothing for missing data
- Regime-switching (Hamilton filter)

Unified framework: one algorithm (the Kalman filter) handles estimation, prediction, and smoothing for all these models.

Missing data: handled naturally — observation equation is simply absent for missing periods.

State Space Representation II

The Kalman Filter: recursions

Initialise: $a_1 = \mathbb{E}[\alpha_1]$, $P_1 = \text{Var}[\alpha_1]$.

Prediction step (at time t , before observing y_t):

$$a_{t|t-1} = T_{t-1}a_{t-1}, \quad P_{t|t-1} = T_{t-1}P_{t-1}T'_{t-1} + R_{t-1}Q_{t-1}R'_{t-1}$$

Updating step (after observing y_t):

$$\begin{aligned} v_t &= y_t - Z_t a_{t|t-1}, \quad F_t = Z_t P_{t|t-1} Z'_t + H_t \\ K_t &= P_{t|t-1} Z'_t F_t^{-1} \quad (\text{Kalman gain}) \\ a_t &= a_{t|t-1} + K_t v_t, \quad P_t = (I - K_t Z_t) P_{t|t-1} \end{aligned}$$

Log-likelihood: $\ell = -\frac{1}{2} \sum_t [\ln |F_t| + v'_t F_t^{-1} v_t]$ — maximise over system parameters.

Section 5

Time-Varying Parameters and Applications

Time-varying parameter (TVP) regression

Observation: $y_t = x_t' \beta_t + \varepsilon_t$, $\varepsilon_t \sim N(0, \sigma^2)$

State: $\beta_{t+1} = \beta_t + \eta_t$, $\eta_t \sim N(0, Q)$

- Parameters β_t follow a random walk: they can drift over time
- $Q = 0$: constant parameter regression (test this restriction)
- Larger Q : parameters vary more rapidly
- Estimated by Kalman filter; Q estimated by MLE or Bayesian methods

Finance: time-varying CAPM beta

CAPM: $r_{i,t} = \alpha_t + \beta_t r_{m,t} + \varepsilon_t$ where β_t evolves as a random walk.

TVP-CAPM captures changes in systematic risk over time — important for:

- Portfolio rebalancing
- Risk management through market cycles
- Measuring effect of corporate events on beta

Marketing: evolving response

TVP advertising response model: $s_t = \alpha_t + \beta_t a_t + \varepsilon_t$
where β_t = advertising elasticity at time t .

Captures:

- Wearout effects (declining β_t)
- Regime shifts after brand changes
- Season- or recession-dependent effectiveness

Pauwels et al. (2002), Korobilis (2013).

Nowcasting with the Kalman filter

Problem: GDP is released quarterly, but we want a monthly estimate (“nowcast”).

Solution: Mixed-frequency state space model

- State: monthly GDP level (unobserved at monthly frequency)
- Observation: quarterly GDP (= average of 3 monthly states), monthly indicators (PMI, retail sales, etc.)
- Kalman filter updates the nowcast as monthly data arrive throughout the quarter
- Used by central banks (ECB, Fed, Banco Central do Brasil) for real-time monitoring

Harvey (1993), Hamilton (1994) Ch. 13. Software: KFAS and dlm in R; statsmodels.tsa.statespace in Python.

Core references

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- Hamilton, J.D. (1994). *Time Series Analysis*. Princeton UP. Chap. 13.
- Bai, J. & Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66, 47–78.
- Andrews, D.W.K. (1993). Tests for parameter instability. *Econometrica*, 61, 821–856.
- Johansen, S. & Nielsen, B. (2016). Asymptotic theory of outlier detection. *Scand. J. Stat.*, 43, 321–348.

Applied papers

- Perron, P. (1989). The great crash, oil price shock, and unit root. *Econometrica*, 57, 1361–1401.
- Korobilis, D. (2013). VAR forecasting using Bayesian variable selection. *JAE*, 28, 204–230.
- Pauwels, K., Hanssens, D.M. & Siddarth, S. (2002). Long-term effects of price promotions. *JMR*, 39, 421–439.
- Hendry, D.F. & Doornik, J.A. (2014). *Empirical Model Discovery*. MIT Press.

Software: R (`strucchange`; `breakpoints`, `Fstats`; `KFAS`; `d1m`) or Python (`statsmodels.tsa.statespace`, `statsmodels.stats.diagnostic`). Applied exercises: Bai-Perron on Brazilian CPI; Kalman TVP-CAPM beta for a Brazilian stock.

Next: Lecture 6

Volatility Modelling: ARCH and GARCH

Stylised facts · GARCH · EGARCH · DCC · Value-at-Risk